

FMS 722.6 TCU — Handoff Note (V14)

This note exists so a fresh session (Claude Code / Opus in VS Code, or any new chat) can pick up with full context: what the firmware IS, where the authoritative docs live, and — most importantly — what is still OPEN after the latest external review (second pass, post-V14 restructure).

Platform: ESP32 DevKit V1, dual-core FreeRTOS, Arduino framework (PlatformIO).

Vehicle: Mercedes 190E (W201), M111 2.3L + Eaton TVS1320 (~1.2 bar), engine on rusEFI.

Transmission: 722.6 Small NAG (W5A330, W203 C230K donor). **Control philosophy:** paddle-commanded shifts + auto-safety (overrev/lug) + coast-down scheduler + kickdown. 2nd-gear garage engagement; 1st is paddle-only. Hard ceiling 6500 rpm.

Where the truth lives

1. `src/Summary_of_knowledge_and_progress.md` (v5.0) — the engineering manual. Mostly current, with three stale spots (see "Doc corrections" below).
2. `Reference/722.6_SHIFT_CLASS_ARCHITECTURE_SPEC.md.pdf` — the ATSG-grounded design spec the V14 phases implemented. Section numbers in code comments refer to it.
3. **Git log** — V14 landed across commits `85bdc69 ... f9420dd` (foundations → phase engine → Adaptation v2 → schedulers → dashboard → EngineProfile → torque recal).
4. This file — status + the prioritized open-items list.

Architecture in one paragraph

Core 1 runs the 1 kHz physics loop (inputs, PCNT kinematics, the **class-parameterised shift phase engine**, safety, standby/garage). Core 0 runs 100 Hz WebSocket telemetry and deferred NVS flushes. Every shift is classified ONCE at `beginShift()` from latched torque (EngineProfile 8×8 RPM×MAP surface, NVS-backed, web-editable) into one of the four ATSG adaptation categories — `SC_POWER_UP / SC_COAST_UP / SC_POWER_DOWN (PD_SPRAG | PD_TIMED) / SC_COAST_DOWN` — and runs that class's profile through `PREP→FILL→TORQUE→INERTIA→LOCK→END` (up) or `PREP→RELEASE→CATCH→LOCK→END` (down). SPC/MPC commands are quantised to 20 ms (ATSG p.80); exit predicates run at 1 kHz. Pressure API is pressure-%: de-energized = 100 = max = OEM "OFF" = in-gear standby = end-of-shift seat. Adaptation v2 trims `{fill_t_cycles ±5, fill_p_trim ±15, apply_trim ±15}` per `[class][shift][torque-bin]`, ATF-gated 60-105 °C, flushed on a 60 s timer or P/N entry.

Verified-correct in the second-pass review (don't re-litigate)

- Class engine, 20 ms quantizer + `_spc_cmd` fractional ramp, PD_SPRAG/PD_TIMED split, coast-down (no sync wait) + decel-delta bind metric, kickdown + money-shift guard, K = 1.641 (tooth-derived), limp ratio-classifier (p.91), standby/garage/crank duties (p.53-54), reverse edge-latch failsafe, mid-shift abort on leaving forward range.
- Adaptation v2: bins captured at initiation, two-sided power-up learning, deadband, mux-protected dirty mask, one-class-per-call flush on Core 0.
- All V13 review fixes present: 10 ms telemetry gate, `char[64]`+seqlock status strings, bounded JSON parse, ingest clamps (`set_cells`)/(`set_profile`).
- Dashboard rework DID land (`get_cells`)/(`set_cells`), Engine Profile tab, 9-phase enum mapping) — the manual's "Phase 9b OPEN" note is stale.

OPEN ITEMS (second-pass review, priority order)

A — fix before bench

1. **TCC not rate-limited (last leftover of review-1 H1).** `updateTCC()` steps $\pm 1...5$ per 1 ms call: 0→85% lock in 85 ms, dump in 17 ms. Quantise TCC to the 20 ms ptick (suggest $\pm 2\%$ /tick lock, $\pm 10\%$ /tick release) and add "hold open until 300 ms after PHASE_END" before resuming lockup control.
2. **Class/torque telemetry not broadcast.** `t_est_nm`, `load_pct`, `shift_class`, `pd_type` are populated but `buildAndSendTelemetryJSON()` never serializes them and `index.html` never shows them. The whole validation plan reads these. Add the four JSON fields + a class/PD badge and load readout on the dashboard.
3. **Y4 garage-vs-shift collision.** `fireShiftSolenoid()` no-ops unless `STATE_OFF`; the garage pulse leaves Y4 `STATE_HOLDING`. A 3→4 shift started inside the 1500 ms lever window silently never happens hydraulically (gear label wrong until limp catches it). Fix: `beginShift()` clears garage ownership when the routing pin is Y4, and let `fireShiftSolenoid()` take over a garage-owned solenoid (kick from holding).

B — correctness / robustness

4. **Ratio feedback is 20 Hz; the detectors assume 1 kHz.** SpeedReader refreshes every 50 ms, so `live_ratio` is frozen between samples → the PD_SPRAG flat check (`(| $\Delta$ ratio| < 0.002)` per 1 ms) is trivially true 49/50 iterations, and INERTIA/CATCH exits carry up to 50 ms latency (~2 pressure ticks of overshoot). Do NOT shorten the PCNT window (20 ms $\Rightarrow$ ± 50 rpm quantization at 60 teeth — worse than the ± 0.05 sync band). Instead: evaluate sync/flat/exit predicates only when a NEW speed sample lands, and compute flatness across consecutive samples.
5. **SC_POWER_DOWN bind learning is dead code.** `bind_detected` is only set in the `SC_COAST_DOWN` branch of PHASE_CATCH, so `learn()`'s power-down path can never fire. Extend the decel-delta metric to PD_TIMED catches, or delete the branch and document it.
6. **Boot pressure state.** `SolenoidDriver::begin()` commands `setLinePressure(0)` / `setShiftPressure(0)` = full coil current + minimum pressures for the whole setup window (SPIFFS + WiFi, ~1-2s, possibly during crank). Boot de-energized (100/100); the scheduler's first standby call takes over.
7. **MPC is a tick late at shift start.** First `applyShiftMPC()` happens on the next 20 ms ptick after the routing solenoid fires. Call it once inside `beginShift()` so line/overlap authority leads the gate (spec PREP intent).

C — calibration / hardware (carried, acknowledged)

8. **Torque model validation.** $K_T=2.43$ / $T_{MAX}=450$ makes full boost $\triangleq$ 100% load (right shape) but 450 Nm is unvalidated — confirm against a datalog/dyno; the SPC apply line and all four torque bins scale from it. Note: W5A330 is rated 330 Nm; running ~450 is 36% over rating — the firm holding-pressure floor is deliberate, keep line authority high at load and treat ATF service as part of the strategy.
9. **ADC linearity.** Still raw `analogRead/4095*3.3`. TPS WOT 2.90 V and the P/N threshold 3.0 V sit in the ESP32 ADC's nonlinear top region — switch to `analogReadMilliVolts()` (one line per read), then re-measure TPS closed/WOT.
10. **Paddles are level-triggered** with 200 ms auto-repeat (hold = repeated shifts). Edge-trigger with debounce if repeat isn't wanted.

11. **M111 crank wheel:** confirm 60 vs 60-2 (missing teeth under-read RPM by ~3% and add jitter through the gap); VR sensor conditioning before PCNT remains a hardware item. `PULSES_PER_REV_ENG` assumes a clean 60.
12. **Hardware rewires already encoded in pins:** MAP → GPIO 33 (ADC1), engine tach → GPIO 23. RP_LOCK polarity and the GPIO15 torque-cut line (strapping pin; default disabled) must be verified before enabling.

D — cosmetics / doc corrections

13. Stale comments: TCU_Data W5A330 header says "K=1.61 applies to both" (value stale AND wrong — K differs per variant: 1.641 small / 1.630 large); same 1.61 in the SpeedReader kinematics comment; `telemetry.shift_phase` comment says "0..7=COMPLETION" vs the 9-entry enum; `crankPulseY3()` clears `_y4_garage_owned` (leftover); `_flare_peak_ratio` tracked but unused; `TPS_VOLTS_CLOSED/WOT` defines in InputManager.h are dead (cal moved to EngineProfile).
 14. **Summary manual (v5.0) corrections:** §2 torque section still shows the old inline $K_T=1.55 / T_{MAX}=330$ — superseded by the EngineProfile 8×8 surface seeded at 2.43/450; §13 "Phase 9b OPEN" is done; §13's "new fields" claim is aspirational until item A-2 lands; §11 TPS cal now lives in EngineProfile, not InputManager.h.
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Validation order (unchanged from spec §11 / manual §14)

Bench: $K=1.641$ TCC-locked ± 20 rpm → scope standby duties + 20 ms SPC quantisation → garage N→D/N→R. Road: coast upshifts → power upshifts 25/50/75% load → coast downshifts to a stop → 3→2 part-throttle (watch the sprag signature: ratio snaps to target with SPC still ~10%) → timed 4→3/5→4 → full kickdown last. Log class/bins/phase durations/SPC/MPC per shift (requires item A-2).